Homework_3

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#PROBLEM 1

```r
cat("Problem 1\n\n")
```

```
## Problem 1
```

```r
library(palmerpenguins)
```

```
##
## Attaching package: 'palmerpenguins'

## The following objects are masked from 'package:datasets':
##
##     penguins, penguins_raw
```

```r
library(ggplot2)

# (a)
cat("\n (a) \n")
```

```
##
##  (a)
```

```r
#Turn the variable species and island into a factor variable.

penguins$species <- as.factor(penguins$species)
penguins$island <- as.factor(penguins$island)
str(penguins[, c("species", "island")])
```

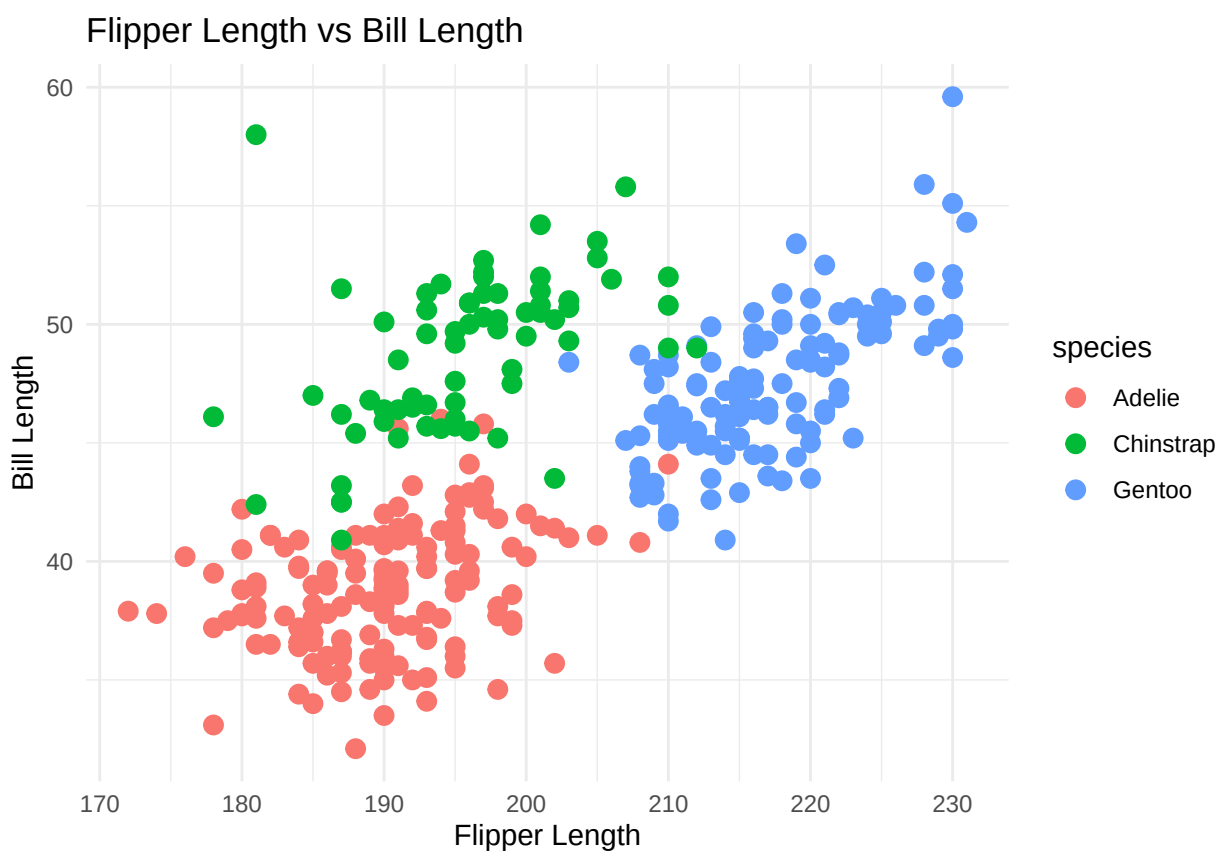
```
## tibble [344 x 2] (S3: tbl_df/tbl/data.frame)
##  $ species: Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ island : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
```

```r
# (b)
cat("\n (b) \n")
```

```
##
##  (b)
```

```r
#Make a scatter plot between flipper length and bill length, indicating species by color

ggplot(penguins, aes(flipper_length_mm, bill_length_mm, color = species)) +
       geom_point(size = 3) +
       labs(title = "Flipper Length vs Bill Length", x = "Flipper Length", y = "Bill Length")+
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# (c)
cat("\n (c) \n")
```

```
##
##  (c)
```

```r
# Compute the Pearson correlation coefficient between flipper length and bill length.

pc <- cor(penguins$flipper_length_mm, penguins$bill_length_mm, method = "pearson", use = "complete.obs")

cat("Pearson correlation coefficient between flipper length and bill length is:", pc)
```

```
## Pearson correlation coefficient between flipper length and bill length is: 0.6561813
```

```r
# (d)
cat("\n (d) \n")
```

```
##
##  (d)
```

```r
# Create three separate data sets by subsetting the penguins dataset in terms of the variable species.

s1 <- subset(penguins, species == "Adelie" )
s2 <- subset(penguins, species == "Gentoo" )
s3 <- subset(penguins, species == "Chinstrap" )
head(s1)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

```r
head(s2)
```

```
## # A tibble: 6 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>           <dbl>         <dbl>             <int>       <int>
## 1 Gentoo  Biscoe           46.1          13.2               211        4500
## 2 Gentoo  Biscoe           50            16.3               230        5700
## 3 Gentoo  Biscoe           48.7          14.1               210        4450
## 4 Gentoo  Biscoe           50            15.2               218        5700
## 5 Gentoo  Biscoe           47.6          14.5               215        5400
## 6 Gentoo  Biscoe           46.5          13.5               210        4550
## # i 2 more variables: sex <fct>, year <int>
```

```r
head(s3)
```

```
## # A tibble: 6 x 8
##   species   island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>     <fct>           <dbl>         <dbl>             <int>       <int>
## 1 Chinstrap Dream            46.5          17.9               192        3500
## 2 Chinstrap Dream            50            19.5               196        3900
## 3 Chinstrap Dream            51.3          19.2               193        3650
## 4 Chinstrap Dream            45.4          18.7               188        3525
## 5 Chinstrap Dream            52.7          19.8               197        3725
## 6 Chinstrap Dream            45.2          17.8               198        3950
## # i 2 more variables: sex <fct>, year <int>
```

```r
# (e)
cat("\n (e) \n")
```

```
##
##  (e)
```

```r
# Compute the Pearson correlation coefficient between flipper length and bill length, separately for ea

pc_1 <- cor(s1$flipper_length_mm, s1$bill_length_mm, method = "pearson", use = "complete.obs")
cat("Pearson correlation coefficient between flipper length and bill length of Adelie is:", pc_1,"\n")
```

```
## Pearson correlation coefficient between flipper length and bill length of Adelie is: 0.3257847
```

```r
pc_2 <- cor(s2$flipper_length_mm, s2$bill_length_mm, method = "pearson", use = "complete.obs")
cat("Pearson correlation coefficient between flipper length and bill length of Gentoo is:", pc_2,"\n")
```

```
## Pearson correlation coefficient between flipper length and bill length of Gentoo is: 0.6611621
```

```r
pc_3 <- cor(s3$flipper_length_mm, s3$bill_length_mm, method = "pearson", use = "complete.obs")
cat("Pearson correlation coefficient between flipper length and bill length of Chinstrap is:", pc_3,"\n
```

```
## Pearson correlation coefficient between flipper length and bill length of Chinstrap is: 0.4716073
```

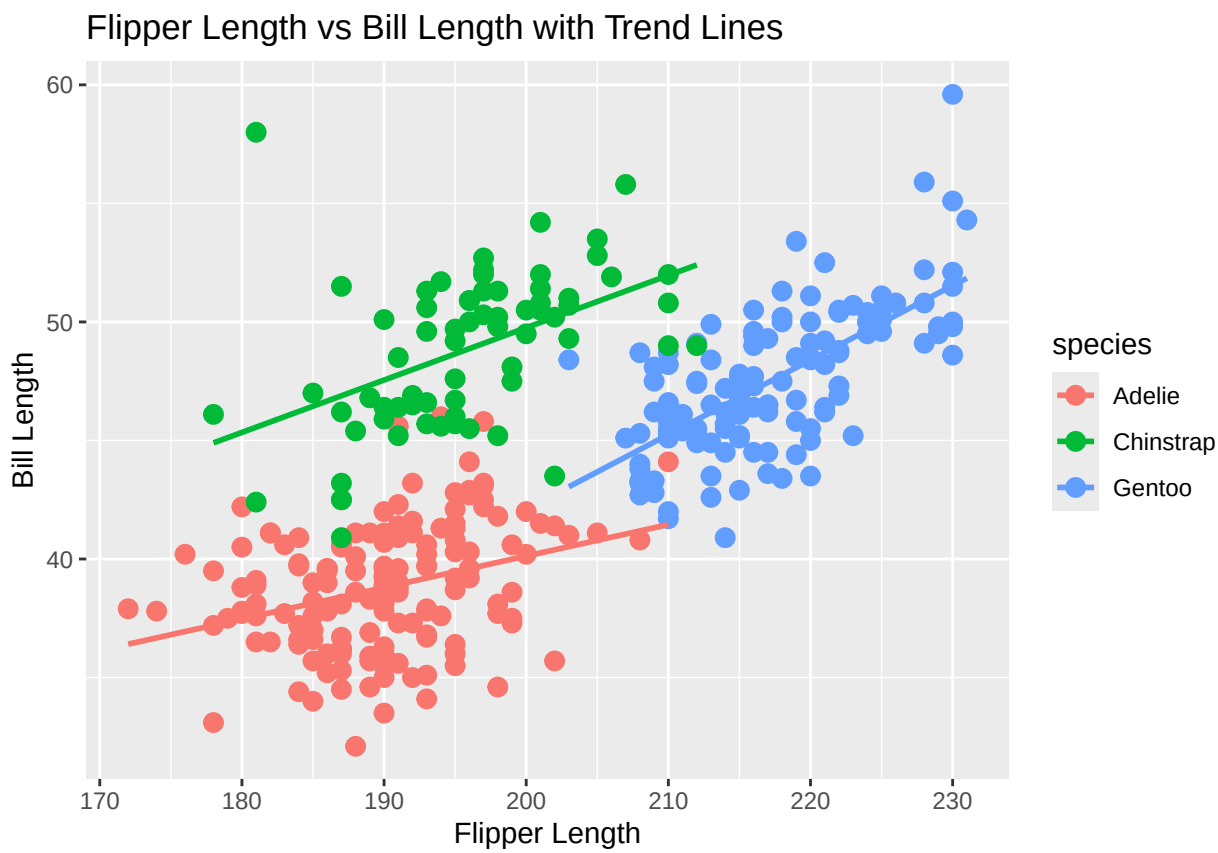
```r
# (f)
cat("\n (f) \n")
```

```
##
##  (f)
```

```r
# Recreate the plot from part (b), adding the trend lines for each species separately

ggplot(penguins, aes(flipper_length_mm, bill_length_mm, color = species)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Flipper Length vs Bill Length with Trend Lines", x = "Flipper Length", y = "Bill Length
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range (`stat_smooth()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# (g)
cat("\n (g) \n")
```

```
##
##  (g)
```

```r
#Find the mean and the median of body mass of penguin in the three islands, and round the answer to two

mean_tor <- round(mean(penguins$body_mass_g[penguins$island == "Torgersen"], na.rm = TRUE),2)

mean_bis <- round(mean(penguins$body_mass_g[penguins$island == "Biscoe"], na.rm = TRUE),2)

mean_dr <- round(mean(penguins$body_mass_g[penguins$island == "Dream"], na.rm = TRUE),2)

median_tor <- round(median(penguins$body_mass_g[penguins$island == "Torgersen"], na.rm = TRUE),2)

median_bis <- round(median(penguins$body_mass_g[penguins$island == "Biscoe"], na.rm = TRUE),2)

median_dr <- round(median(penguins$body_mass_g[penguins$island == "Dream"], na.rm = TRUE),2)

cat("The mean  body mass of penguin in the Torgersen island is",mean_tor,", and the median body mass is
```

```
## The mean  body mass of penguin in the Torgersen island is 3706.37 , and the median body mass is 3700
```

```r
cat("The mean  body mass of penguin in the Biscoe island is",mean_bis,", and the median body mass is",me
```

```
## The mean  body mass of penguin in the Biscoe island is 4716.02 , and the median body mass is 4775 .
```

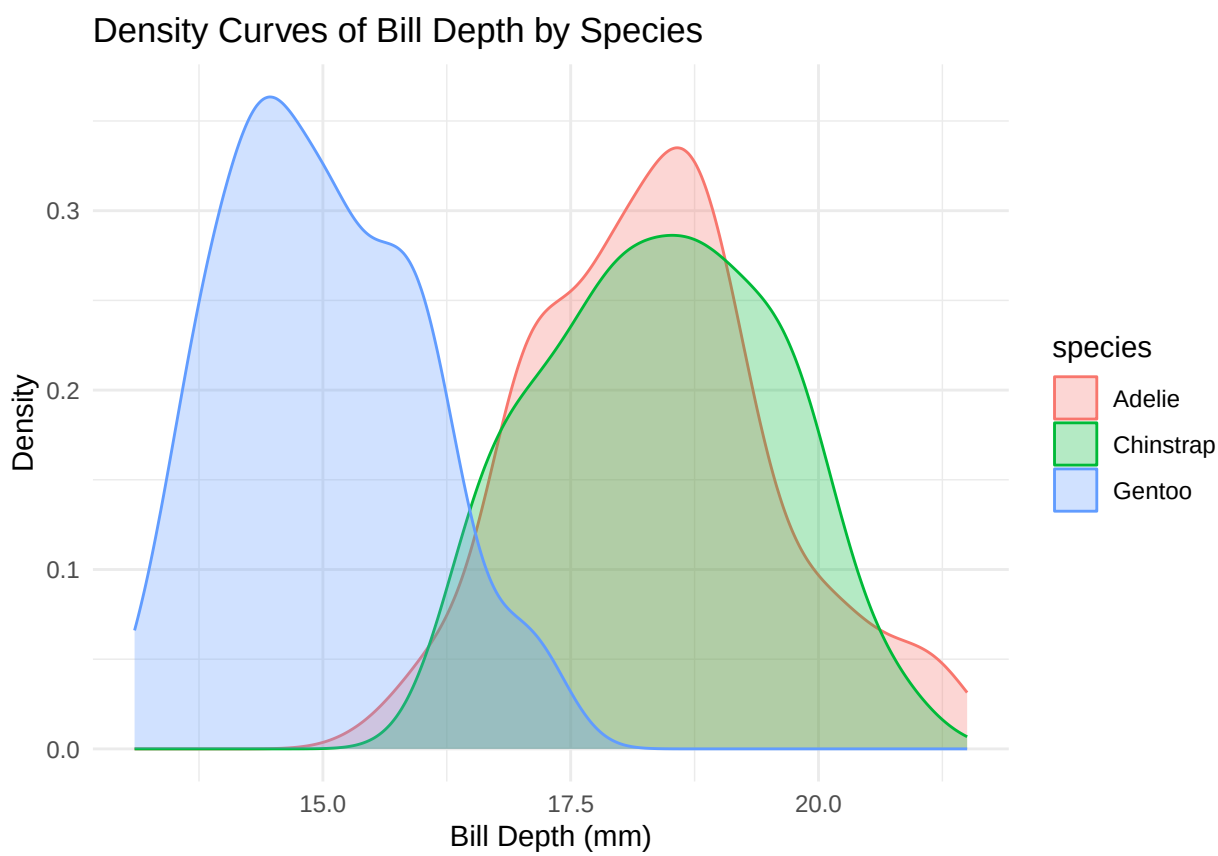
```r
cat("The mean  body mass of penguin in the Dream island is",mean_dr,", and the median body mass is",medi
```

```
## The mean  body mass of penguin in the Dream island is 3712.9 , and the median body mass is 3687.5 .
```

```r
# (h)
cat("\n (h) \n")
```

```
##
##  (h)
```

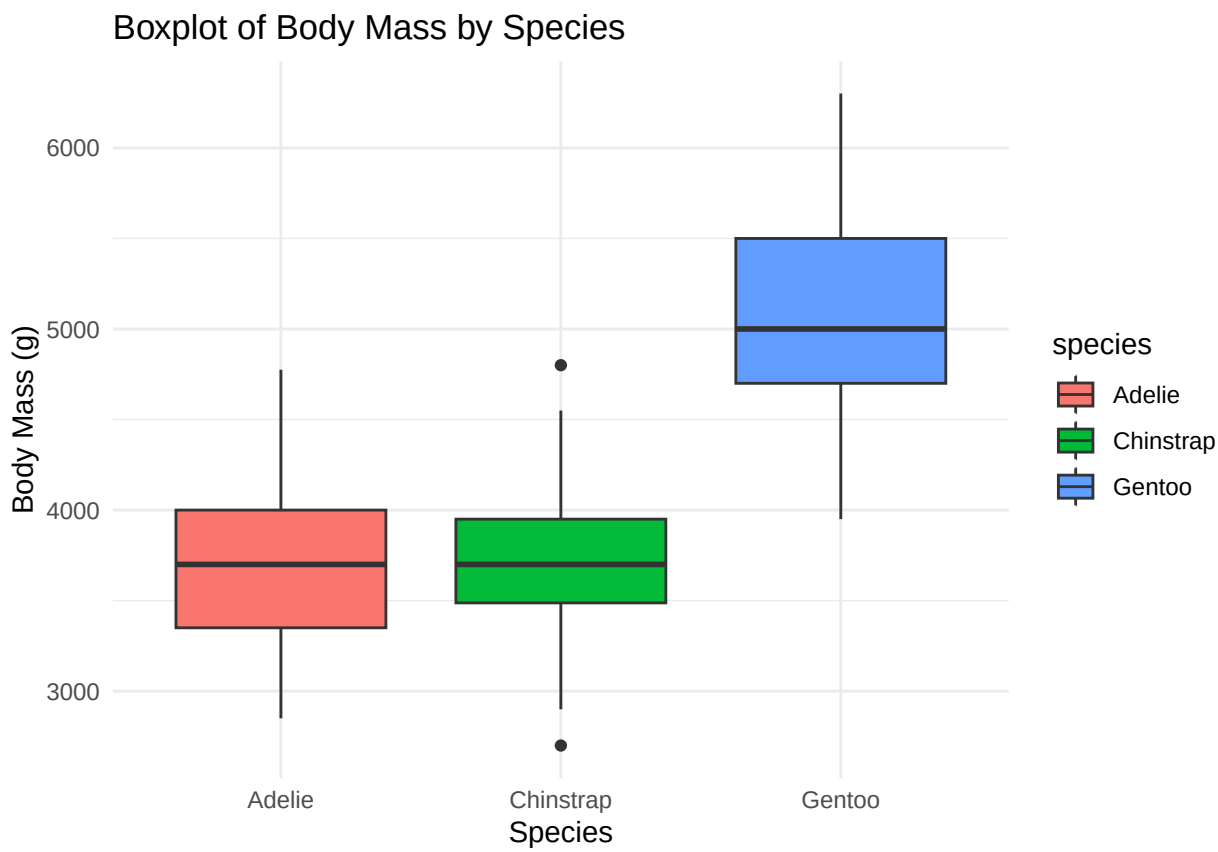
```r
# Make density curves of the bill depth for all the three species in one plot

ggplot(penguins, aes(x = bill_depth_mm, color = species, fill = species)) +
  geom_density(alpha = 0.3, na.rm = TRUE) +
  labs(title = "Density Curves of Bill Depth by Species", x = "Bill Depth (mm)", y = "Density")+
  theme_minimal()
```



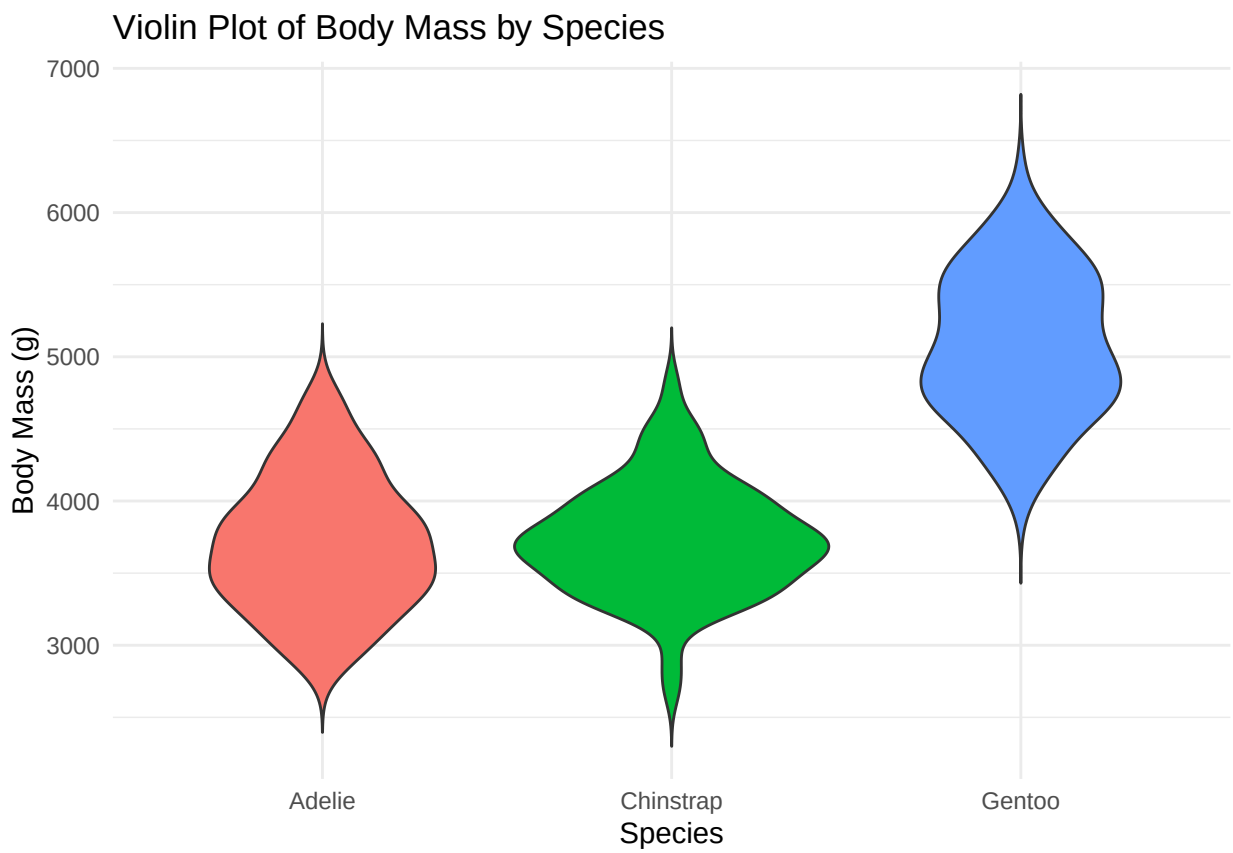
```r
# (i)
cat("\n (i) \n")
```

```
##
##  (i)
```

```r
# Make a boxplot and a violin plot of the body mass for all the three species, respectively.

ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Boxplot of Body Mass by Species", x = "Species", y = "Body Mass (g)") +
  theme_minimal()
```



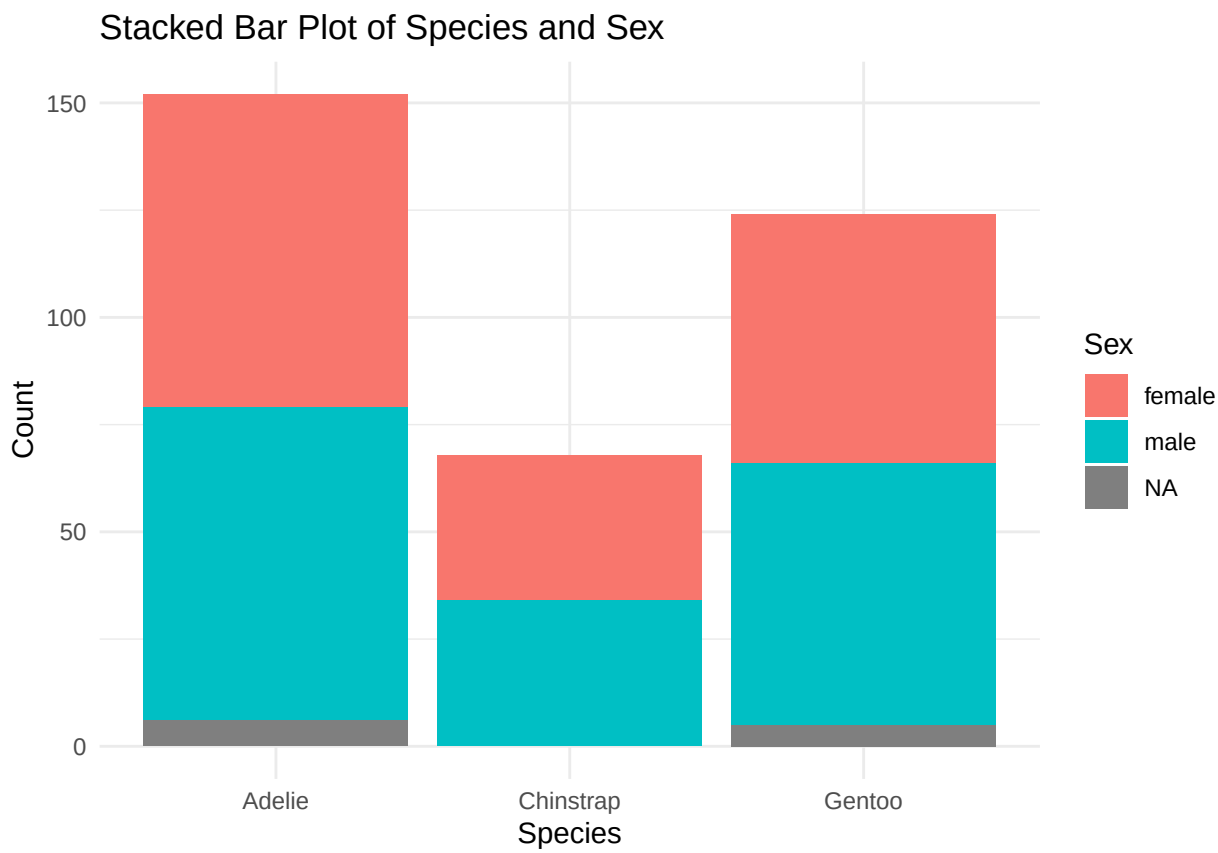

```r
ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +
  geom_violin(na.rm = TRUE, trim = FALSE) +
  labs(title = "Violin Plot of Body Mass by Species", x = "Species", y = "Body Mass (g)") +
  theme_minimal() +
  theme(legend.position = "none")
```




```r
# (j)
cat("\n (j) \n")
```

```
##
##  (j)
```

```r
# Make a stacked bar plot to show the relationship between species and sex variables.

ggplot(penguins, aes(x = species, fill = sex)) +
  geom_bar(na.rm = TRUE) +
  labs(title = "Stacked Bar Plot of Species and Sex", x = "Species", y = "Count",fill = "Sex") +
  theme_minimal()
```



#PROBLEM 2

```
cat("Problem 2\n\n")
```

```
## Problem 2
```

```
heart <- read.csv("heart.csv", stringsAsFactors = FALSE)
str(heart)
```

```
## 'data.frame':    1025 obs. of  14 variables:
##  $ age     : int  52 53 70 61 62 58 58 55 46 54 ...
##  $ sex     : int  1 1 1 1 0 0 1 1 1 1 ...
##  $ cp      : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ trestbps: int  125 140 145 148 138 100 114 160 120 122 ...
##  $ chol    : int  212 203 174 203 294 248 318 289 249 286 ...
##  $ fbs     : int  0 1 0 0 1 0 0 0 0 0 ...
##  $ restecg : int  1 0 1 1 1 0 2 0 0 0 ...
##  $ thalach : int  168 155 125 161 106 122 140 145 144 116 ...
##  $ exang   : int  0 1 1 0 0 0 0 1 0 1 ...
##  $ oldpeak : num  1 3.1 2.6 0 1.9 1 4.4 0.8 0.8 3.2 ...
##  $ slope   : int  2 0 0 2 1 1 0 1 2 1 ...
##  $ ca      : int  2 0 0 1 3 0 3 1 0 2 ...
##  $ thal    : int  3 3 3 3 2 2 1 3 3 2 ...
##  $ target  : int  0 0 0 0 0 1 0 0 0 0 ...
```

```r
# (a)
cat("\n (a) \n")
```

```
##
##  (a)
```

```r
# Turn the variable thal into a factor variable
unique(heart$thal)
```

```
## [1] 3 2 1 0
```

```r
heart$thal <- as.factor(heart$thal)

str(heart[, "thal"])
```

```
##  Factor w/ 4 levels "0","1","2","3": 4 4 4 4 3 3 2 4 4 3 ...
```

```r
# (b)
cat("\n (b) \n")
```

```
##
##  (b)
```

```r
# turn the variable cp into a factor variable with four labels: "Typ", "Aty", "Non", and "Asy".

unique(heart$cp)
```

```
## [1] 0 1 2 3
```

```r
heart$cp <- factor( heart$cp, levels = c(0,1,2,3), labels = c("Typ", "Aty", "Non", "Asy"))

str(heart[, "cp"])
```

```
##  Factor w/ 4 levels "Typ","Aty","Non",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```r
# (c)
cat("\n (c) \n")
```

```
##
##  (c)
```

```r
# Create a new factor variable cholesterol level based on cholesterol levels for adults, where levels b

heart$chol_level <- cut(
  heart$chol,
  breaks <- c(-Inf, 200, 240, Inf),
  labels <- c("Desirable", "Medium", "High"),
  right <- FALSE
)

str(heart[, "chol_level"])
```

```
##  Factor w/ 3 levels "Desirable","Medium",..: 2 2 1 2 3 3 3 3 3 3 ...
```

```r
# (d)
cat("\n (d) \n")
```

```
##
##  (d)
```

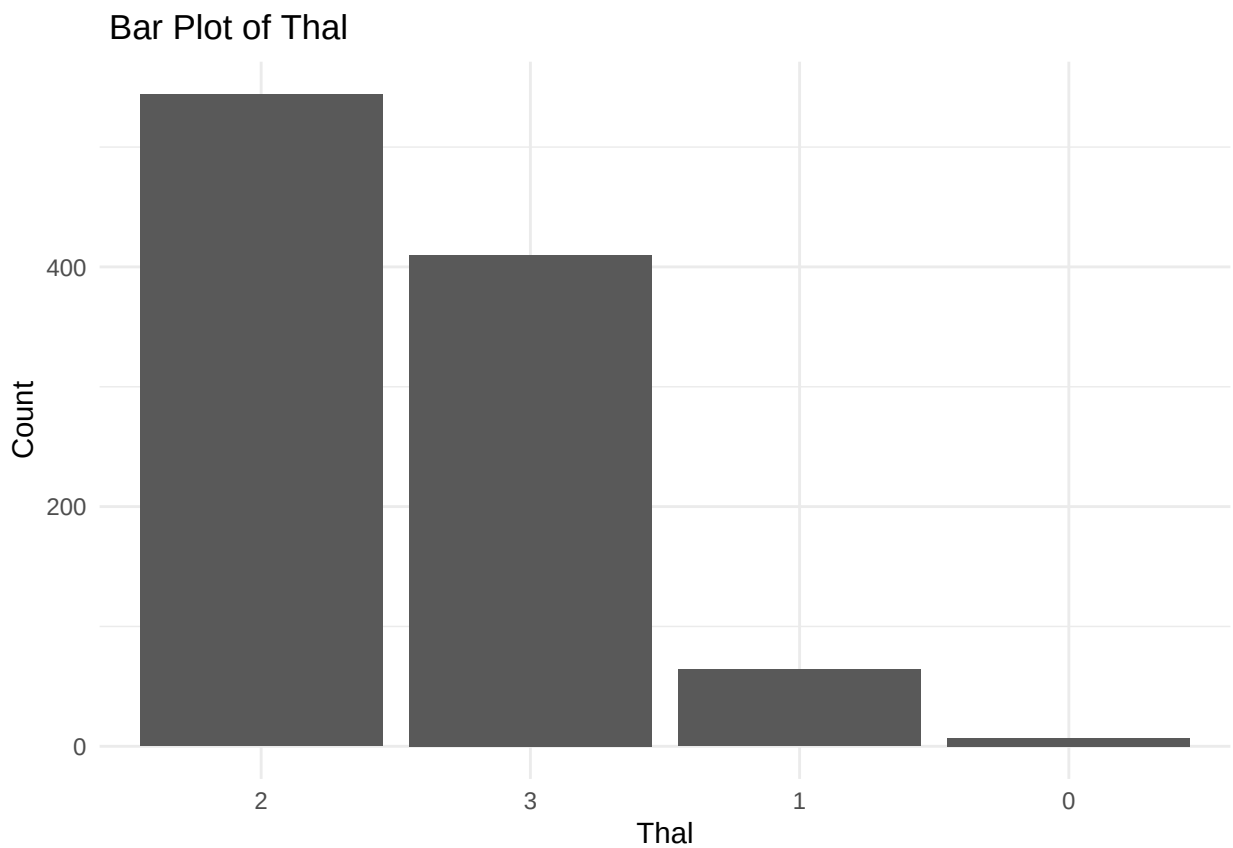
```r
# (d) Create a new factor variable blood level based on resting blood pressure levels for adults, where

heart$blood_level <- cut(
  heart$trestbps,
  breaks <- c(-Inf, 120, 139, Inf),
  labels <- c("Normal", "At Risk", "High" ),
  right <- FALSE
)

str(heart[, "blood_level"])
```

```
##  Factor w/ 3 levels "Normal","At Risk",..: 2 3 3 3 2 1 1 3 1 2 ...
```

```r
# (e)
cat("\n (e) \n")
```

```
##
##  (e)
```

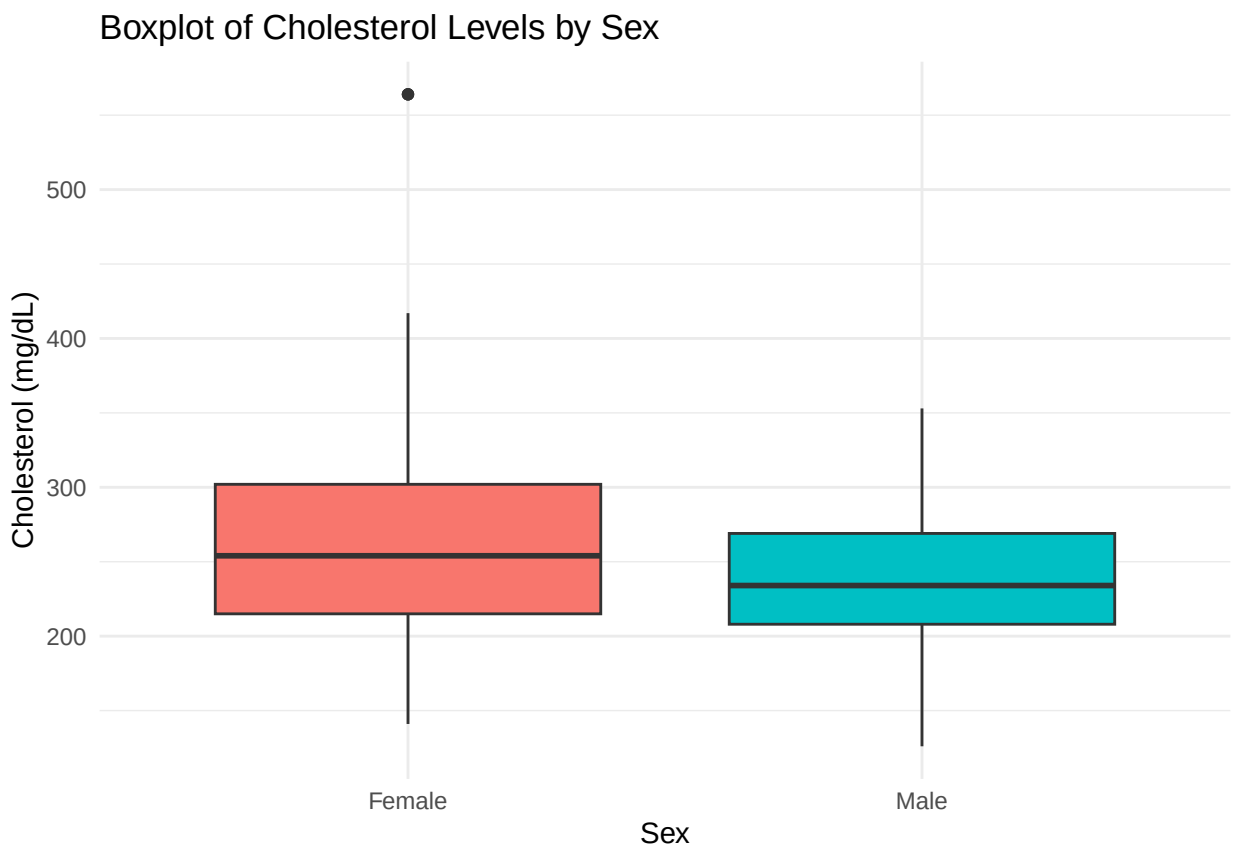
```r
# Draw a bar plot of the variable thal and arrange the bars in decreasing order in terms of the number


ggplot(heart, aes(x= reorder(thal, -table(thal)[thal])))+
  geom_bar(na.rm = TRUE)+
  labs(title = " Bar Plot of Thal", x ="Thal", y= "Count")+
  theme_minimal()
```




```
# (f)
cat("\n (f) \n")
```

```
##
##  (f)
```

```
# Make a boxplot of cholesterol levels for female and males patients, respectively, and report their me
heart$sex <- factor(heart$sex,
                    levels = c(0,1),
                    labels = c("Female", "Male"))

ggplot(heart, aes(x = sex, y = chol, fill = sex)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Boxplot of Cholesterol Levels by Sex", x = "Sex", y = "Cholesterol (mg/dL)") +
  theme_minimal() +
  theme(legend.position = "none")
```




```
mean_chol_fe <- round(mean(heart$chol[heart$sex == "Female"], na.rm = TRUE), 2)
mean_chol_ma <- round(mean(heart$chol[heart$sex == "Male"], na.rm = TRUE), 2)

median_chol_fe <- round(median(heart$chol[heart$sex == "Female"], na.rm = TRUE), 2)
median_chol_ma <- round(median(heart$chol[heart$sex == "Male"], na.rm = TRUE), 2)

sd_chol_fe <- round(sd(heart$chol[heart$sex == "Female"], na.rm = TRUE), 2)
sd_chol_ma <- round(sd(heart$chol[heart$sex == "Male"], na.rm = TRUE), 2)

IQR_chol_fe <- round(IQR(heart$chol[heart$sex == "Female"], na.rm = TRUE), 2)
IQR_chol_ma <- round(IQR(heart$chol[heart$sex == "Male"], na.rm = TRUE), 2)

cat("Female Patients:\n")
```

```
## Female Patients:
```

```
cat("Mean:", mean_chol_fe, "\n")
```

```
## Mean: 261.46
```

```
cat("Median:", median_chol_fe, "\n")
```

```
## Median: 254
```

```r
cat("Standard Deviation:", sd_chol_fe, "\n")
```

```
## Standard Deviation: 64.47
```

```r
cat("IQR:", IQR_chol_fe, "\n\n")
```

```
## IQR: 87
```

```r
cat("Male Patients:\n")
```

```
## Male Patients:
```

```r
cat("Mean:", mean_chol_ma, "\n")
```

```
## Mean: 239.24
```

```r
cat("Median:", median_chol_ma, "\n")
```

```
## Median: 234
```

```r
cat("Standard Deviation:", sd_chol_ma, "\n")
```

```
## Standard Deviation: 43.16
```

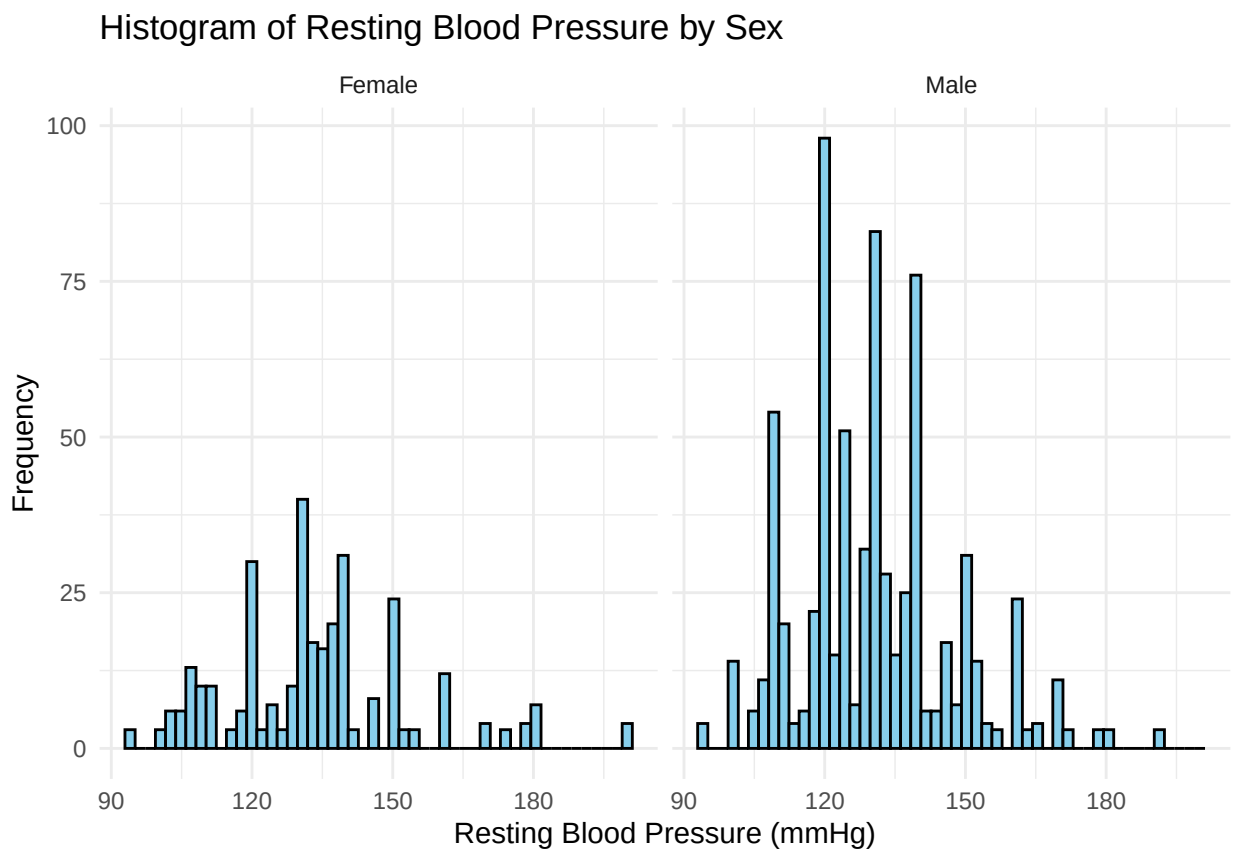
```r
cat("IQR:", IQR_chol_ma, "\n")
```

```
## IQR: 61
```

```r
# (g)
cat("\n (g) \n")
```

```
##
##  (g)
```

```r
# Make a histogram of resting blood pressures for female and male patients, respectively, and describe

ggplot(heart, aes(x = trestbps)) +
  geom_histogram(bins = 50, fill = "skyblue", color = "black", na.rm = TRUE) +
  facet_wrap(~ sex) +
  labs(title = "Histogram of Resting Blood Pressure by Sex",
       x = "Resting Blood Pressure (mmHg)",
       y = "Frequency") +
  theme_minimal()
```



#PROBLEM 3

```
cat("Problem 3\n\n")
```

```
## Problem 3
```

```
head(diamonds)
```

```
## # A tibble: 6 x 10
##   carat cut       color clarity depth table price     x     y     z
##   <dbl> <ord>     <ord> <ord>   <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal     E     SI2      61.5    55   326  3.95  3.98  2.43
## 2  0.21 Premium   E     SI1      59.8    61   326  3.89  3.84  2.31
## 3  0.23 Good      E     VS1      56.9    65   327  4.05  4.07  2.31
## 4  0.29 Premium   I     VS2      62.4    58   334  4.2   4.23  2.63
## 5  0.31 Good      J     SI2      63.3    58   335  4.34  4.35  2.75
## 6  0.24 Very Good J     VVS2     62.8    57   336  3.94  3.96  2.48
```

```
# (a)
cat("\n (a) \n")
```

```
##
##  (a)
```

```r
# Turn the variables cut, color, clarity into factor variables

diamonds$cut <- as.factor(diamonds$cut)
diamonds$color <- as.factor(diamonds$color)
diamonds$clarity <- as.factor(diamonds$clarity)

str(diamonds[, c("cut", "color", "clarity")])
```

```
## tibble [53,940 x 3] (S3: tbl_df/tbl/data.frame)
##  $ cut    : Ord.factor w/ 5 levels "Fair"<"Good"<..: 5 4 2 4 2 3 3 3 1 3 ...
##  $ color  : Ord.factor w/ 7 levels "D"<"E"<"F"<"G"<..: 2 2 2 6 7 7 6 5 2 5 ...
##  $ clarity: Ord.factor w/ 8 levels "I1"<"SI2"<"SI1"<..: 2 3 5 4 2 6 7 3 4 5 ...
```

```r
# (b)
cat("\n (b) \n")
```

```
##
##  (b)
```

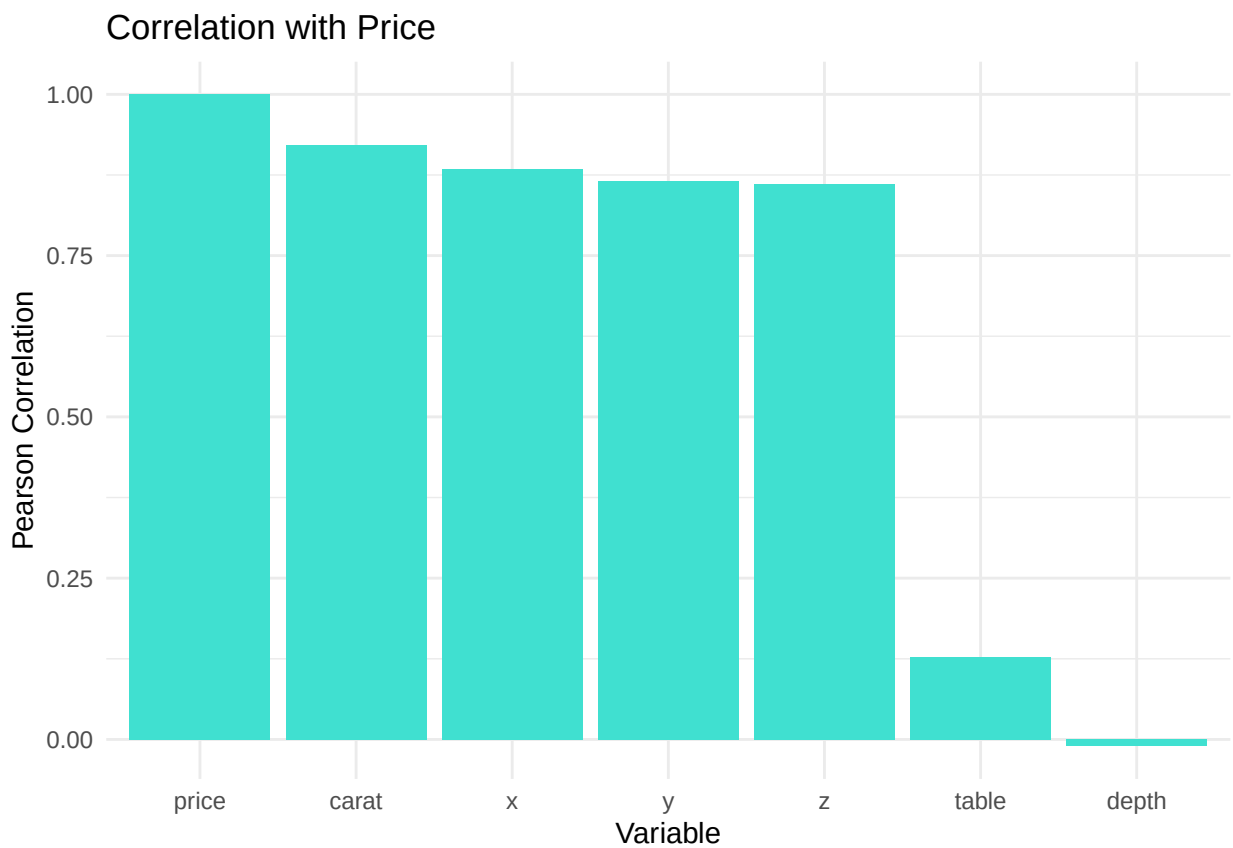
```r
# Calculate the Pearson correlation coefficient between price and each of the variables carat, depth, t

vars <- c("carat", "depth", "table", "price", "x", "y", "z")

cor_results <- sapply(vars, function(v) {
  cor(diamonds$price, diamonds[[v]], method = "pearson")
})

cat("Pearson correlation coefficient between price and each of the variables carat, depth, table, price
```

```
## Pearson correlation coefficient between price and each of the variables carat, depth, table, price, 
##
```

```r
cor_results
```

```
##      carat      depth      table      price          x          y          z
##  0.9215913 -0.0106474  0.1271339  1.0000000  0.8844352  0.8654209  0.8612494
```

```r
# (c)
# Make a bar plot of the resulting correlations in part (b) and arrange the variables in decreasing ord

cor_df <- data.frame(variable = names(cor_results), correlation = cor_results)

ggplot(cor_df,aes(x = reorder(variable, -correlation), y = correlation)) +
  geom_bar(stat = "identity", fill = "turquoise") +
  labs(title = "Correlation with Price",x = "Variable",y = "Pearson Correlation") +
  theme_minimal()
```




```r
# (d)
cat("\n (d) \n")
```

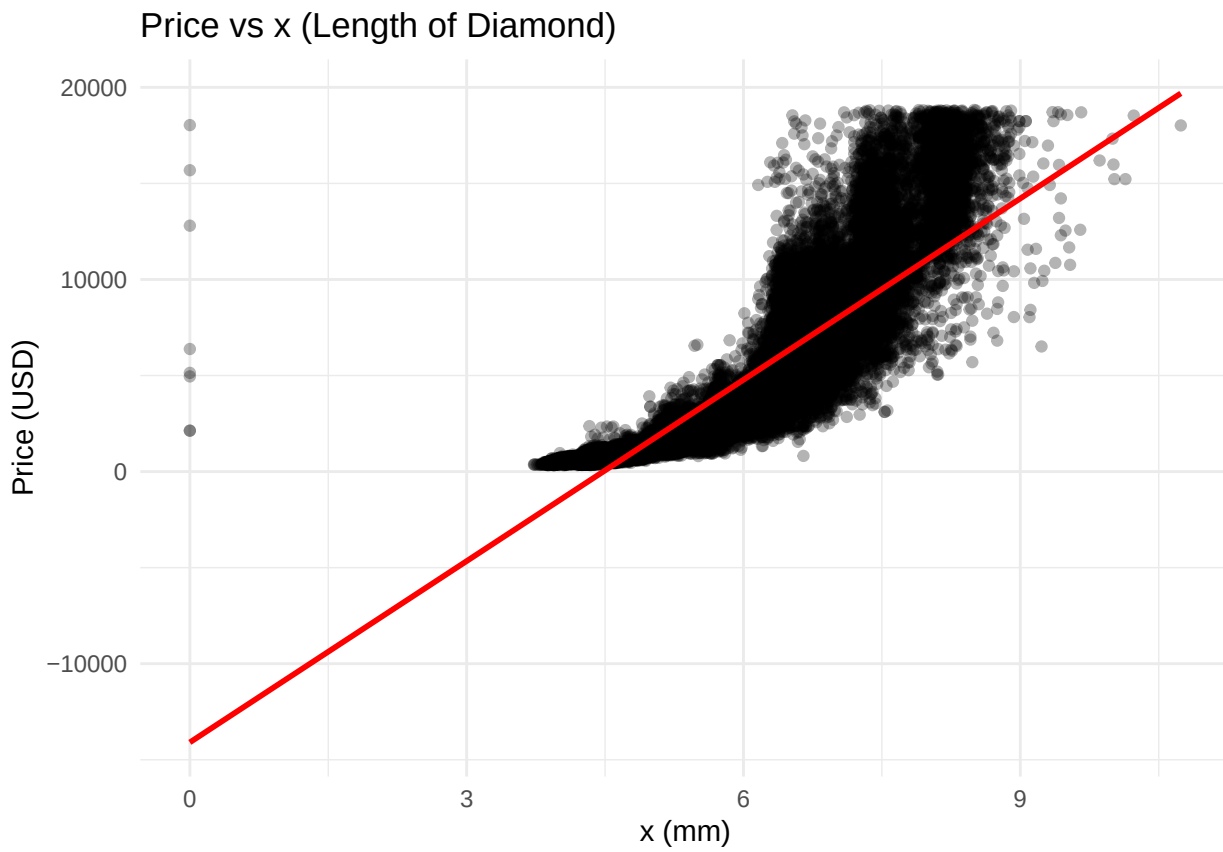
```
##
##  (d)
```

```r
# Make a scatter plot where y-axis represents price and x-axis represents the variable with the largest

vars2 <- c("depth", "table", "price", "x", "y", "z")

cor_carat <- sapply(vars2, function(v) {
  cor(diamonds$carat, diamonds[[v]], method = "pearson")
})

round(cor_carat, 4)
```

```
##  depth  table  price      x      y      z
## 0.0282 0.1816 0.9216 0.9751 0.9517 0.9534
```

```r
ggplot(diamonds, aes(x = x, y = price)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Price vs x (Length of Diamond)", x = "x (mm)", y = "Price (USD)") +
  theme_minimal()
```

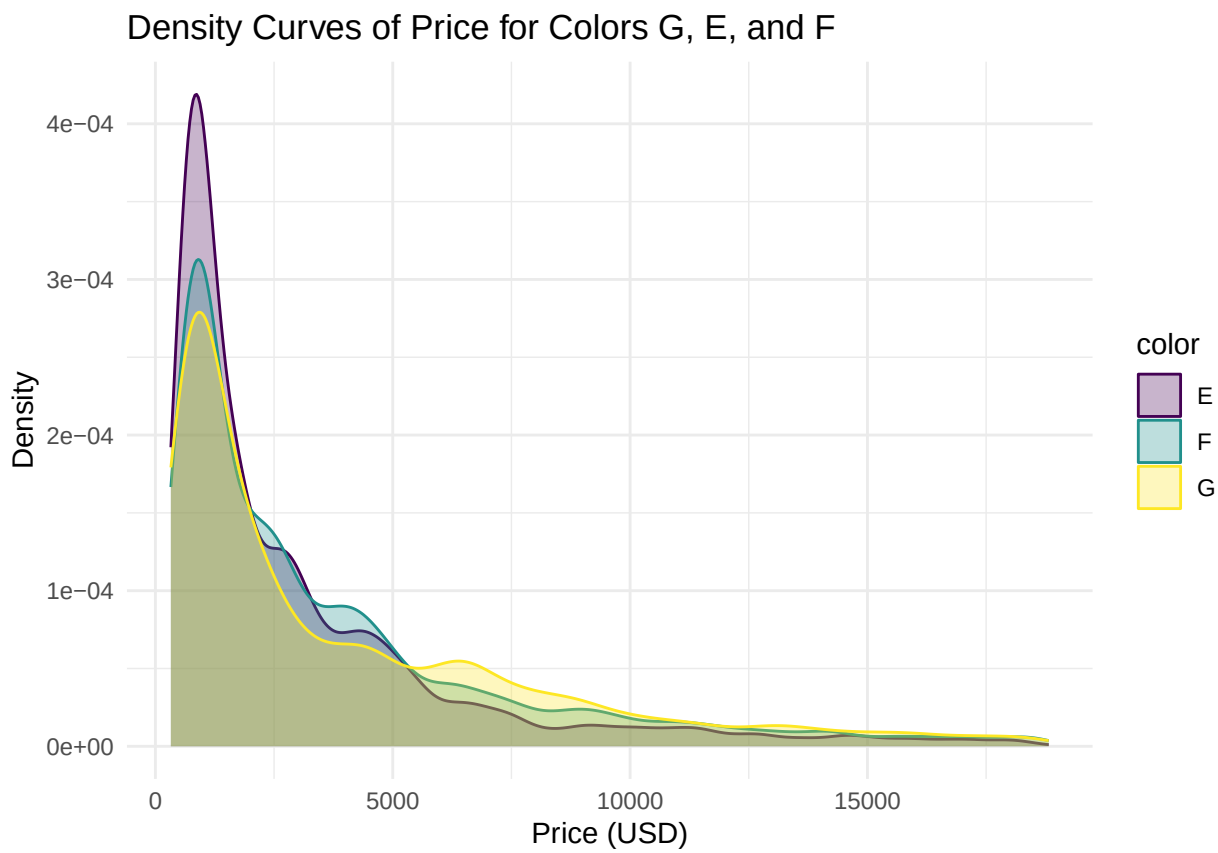
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# (e)
cat("\n (e) \n")
```

```
##
##  (e)
```

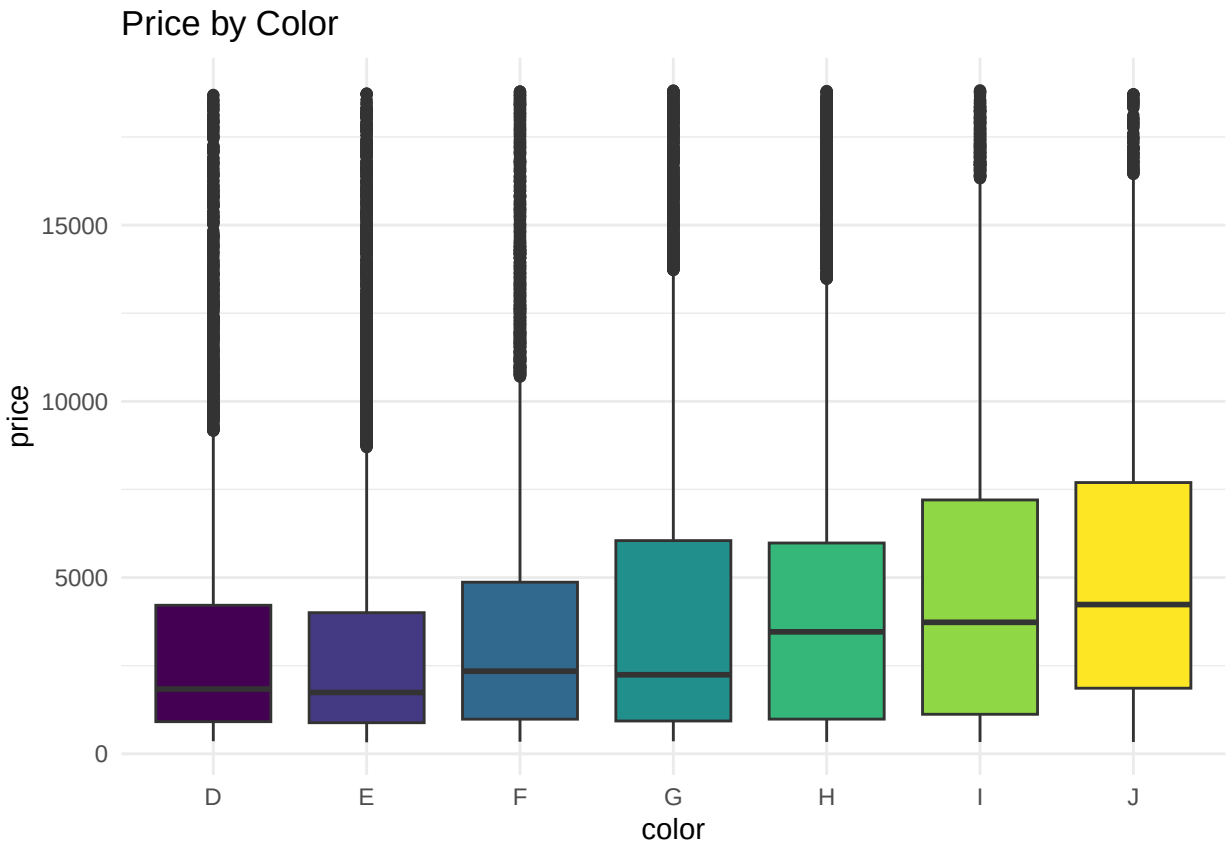
```r
# The main colors of the diamonds are G, E and F, since they have larger numbers of samples than any ot

dim_sub <- subset(diamonds, color %in% c("G", "E", "F"))

ggplot(dim_sub, aes(x = price, color = color, fill = color)) +
  geom_density(alpha = 0.3) +
  labs(title = "Density Curves of Price for Colors G, E, and F", x = "Price (USD)",y =     "Density") +
  theme_minimal()
```



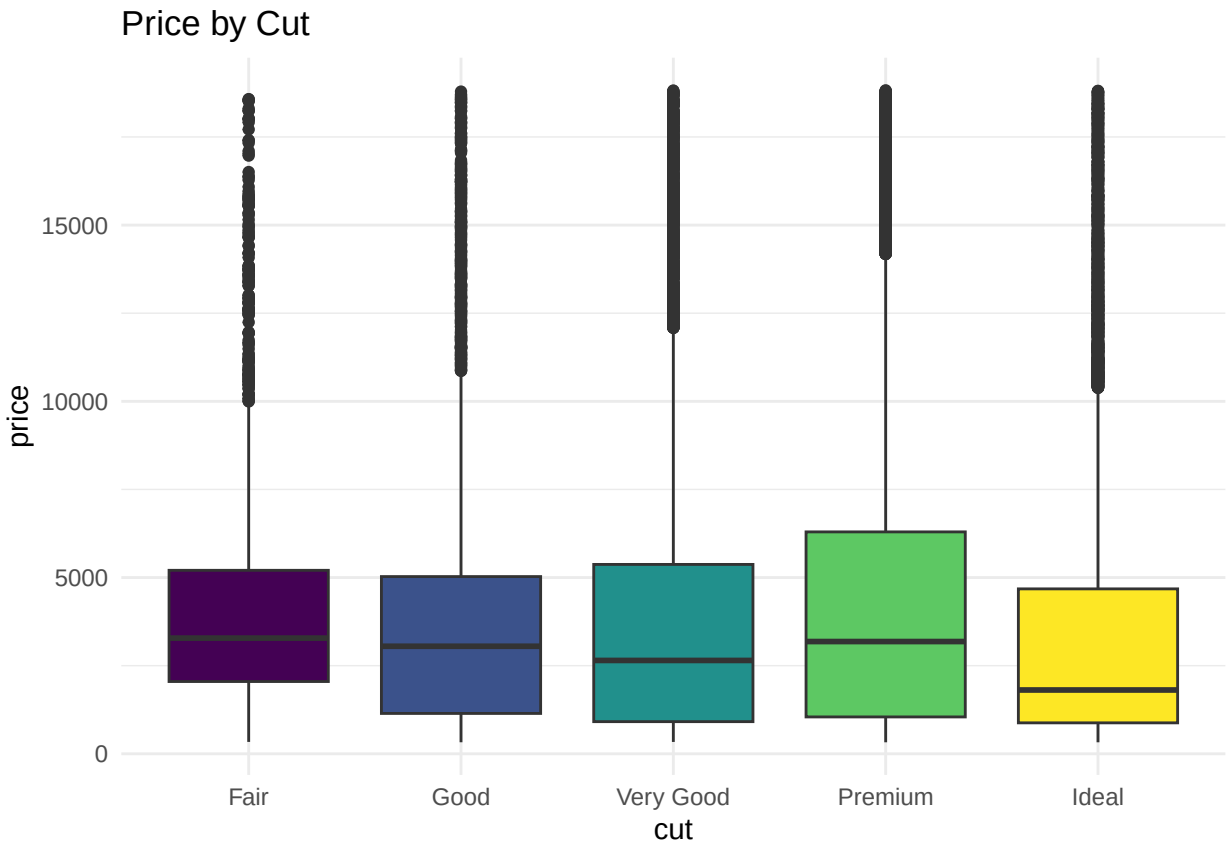

```r
# (f)
cat("\n (f) \n")
```

```
##
##  (f)
```

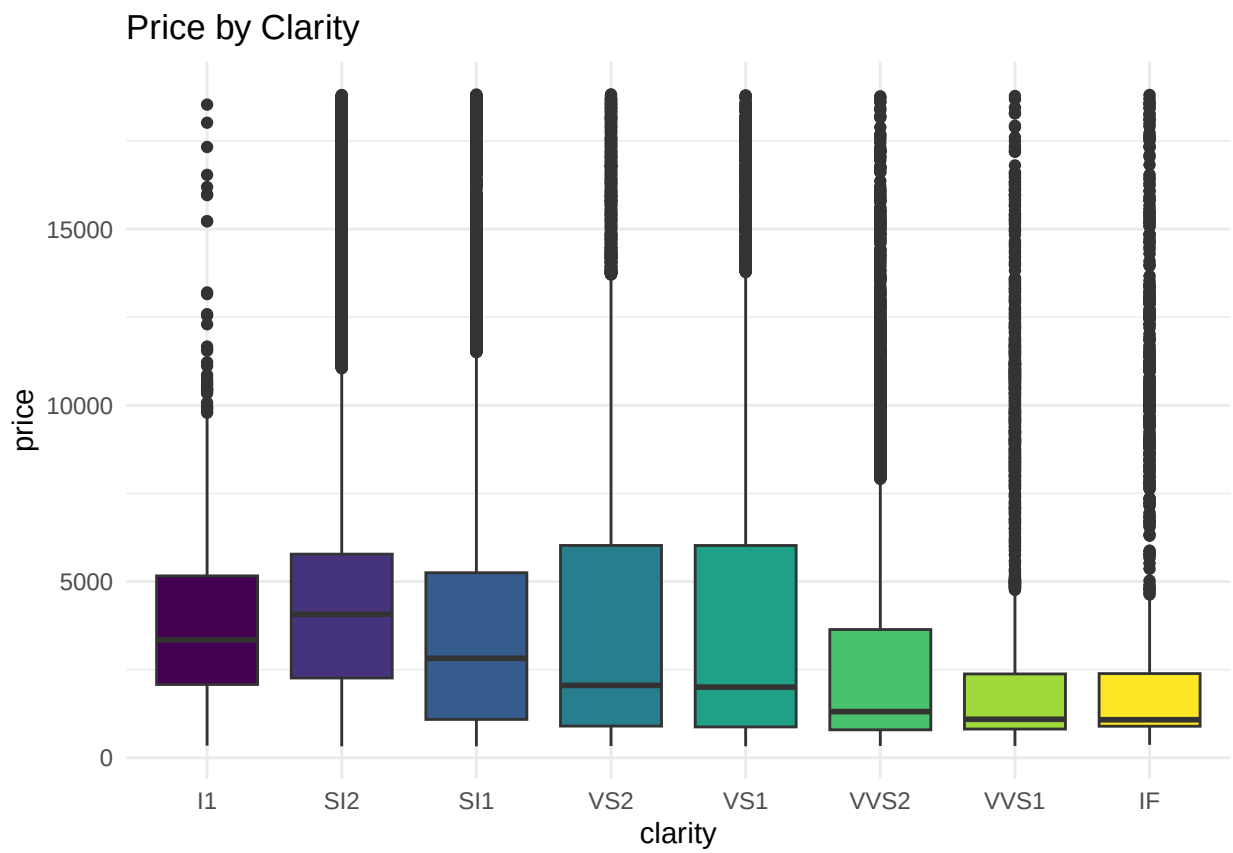
```r
# Make three side-by-side box plots to compare price of diamond among the categorical variable, includi

ggplot(diamonds, aes(x = color, y = price, fill = color)) +
  geom_boxplot() +
  theme_minimal() +
  theme(legend.position = "none") +
  labs(title = "Price by Color")
```




```r
ggplot(diamonds, aes(x = cut, y = price, fill = cut)) +
  geom_boxplot() +
  theme_minimal() +
  theme(legend.position = "none") +
  labs(title = "Price by Cut")
```



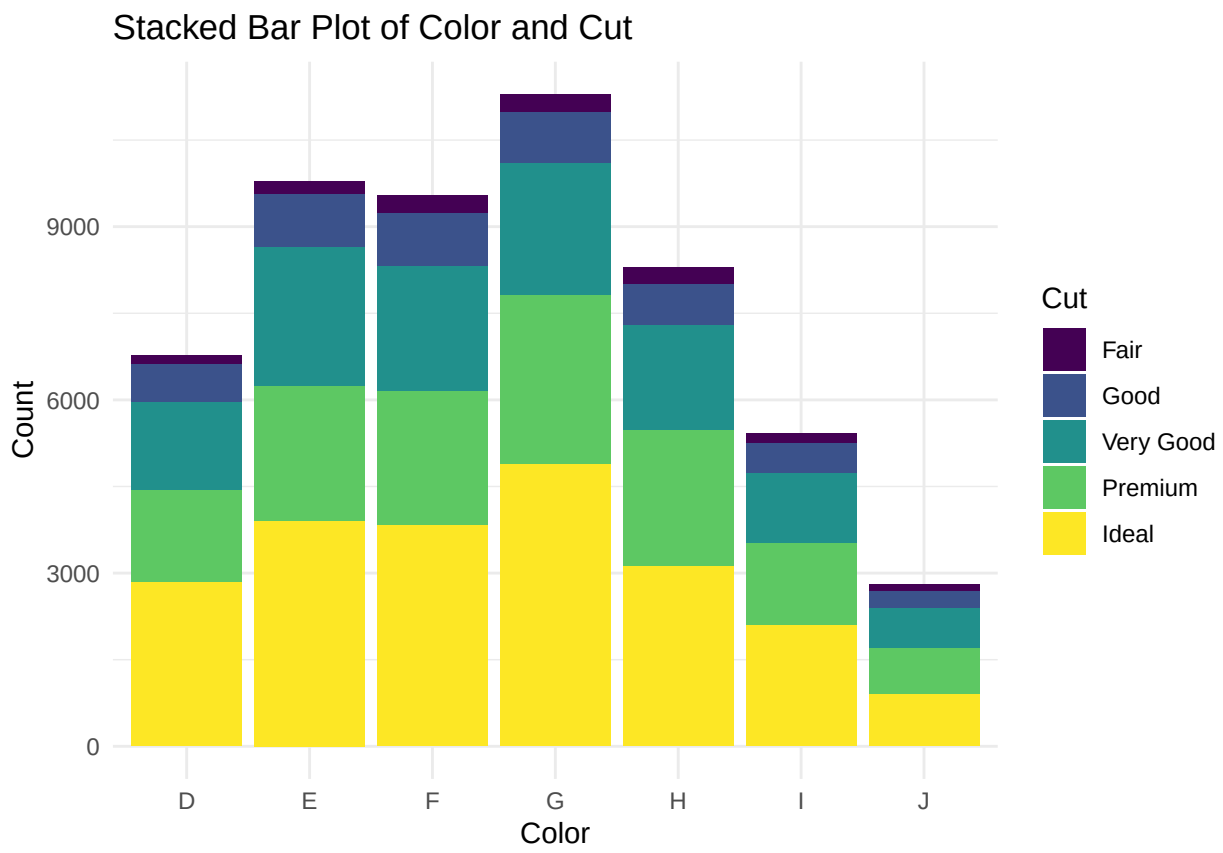

```r
ggplot(diamonds, aes(x = clarity, y = price, fill = clarity)) +
  geom_boxplot() +
  theme_minimal() +
  theme(legend.position = "none") +
  labs(title = "Price by Clarity")
```




```r
# (g)
cat("\n (g) \n")
```

```
##
##  (g)
```

```r
# Make a stacked bar plot to show the relationship between color and cut.

ggplot(diamonds, aes(x = color, fill = cut)) +
  geom_bar(position = "stack") +
  labs(title = "Stacked Bar Plot of Color and Cut", x = "Color", y = "Count", fill = "Cut") +
  theme_minimal()
```



(d) As x increases, price increases.

(e) The density curves for colors G, E, and F are all right-skewed and unimodal. Most diamond prices are concentrated in the lower price range, with a long tail extending toward higher prices. Although the general shape of the distributions is similar across the three colors, slight differences in concentration and spread are observed, with color E showing a higher peak at lower prices